Category theory: definition and first examples

Notes from Carlos' lessons

Source lesson: 2026-07-22 (category theory).

This lesson is a definitions-and-examples board; there are no long proofs. Below are the definitions and, for each example, the small verification that it really is a category.

Definition

Definition 1. A **category** consists of

1. a collection of **objects** $(A, B, C, \dots)$;
2. a collection of **morphisms (arrows)** $f : A \to B$ between objects;
3. for every object X an **identity** arrow $\mathrm{id}_X : X \to X$;
4. **composition**: for $f : A \to B$ and $g : B \to C$ an arrow $g \circ f : A \to C$;
5. laws: $f \circ \mathrm{id}_A = f$, $\mathrm{id}_B \circ f = f$, and composition is associative, $h \circ (g \circ f) = (h \circ g) \circ f$.

(The board's picture: $f : A \to B$, $g : B \to C$, $g \circ f : A \to C$, loops id_A and id_B; and the check $f(\mathrm{id}_A(t)) = f(t)$ for functions.)

Examples from the board

1. **Set.** Objects: sets. Arrows: functions. Identity: $x \mapsto x$. Composition of functions is associative and has the identity laws, as checked pointwise.

2. **Poset (or preorder) category.** Take a partially ordered set; objects are its elements, and there is *exactly one* arrow $a \to b$ iff $a \le b$. Identities exist by reflexivity ($a \le a$); composition $(b \le c) \circ (a \le b) = (a \le c)$ exists by transitivity; the laws hold automatically since there is at most one arrow between any two objects. Example drawn: $\cdots \to -2 \to -1 \to 0 \to 1 \to 2 \to 3 \to \cdots$ with identity loops ($1 \le 3$ is the composite $1 \to 2 \to 3$).

   - A *partial order* requires antisymmetry ($a \le b$ and $b \le a$ imply $a = b$).
   - A *preorder* requires only reflexivity and transitivity. The board's example is **divisibility**: $n \mid n$, and $a \mid b,\ b \mid c \Rightarrow a \mid c$. On the positive integers it is a partial order; on $\mathbb{Z}$ it is only a preorder, since $a \mid -a$ and $-a \mid a$ with $a \neq -a$ ("perhaps $a = -b$"). The drawn diagram of Div: $1 \to 2 \to 4 \to \cdots$ and $1 \to 3 \to \cdots$.

3. **Grp.** Objects: groups. Arrows: group homomorphisms $\varphi : G \to H$ with $\varphi(ab) = \varphi(a)\varphi(b)$ (hence $\varphi(e_G) = e_H$ and $\varphi(g^{-1}) = \varphi(g)^{-1}$; see the topic *Group theory basics*). The composite

of homomorphisms is a homomorphism, so this is a category. Example: $\mathbb{Z} \xrightarrow{\text{mod } n} \mathbb{Z}_n$ is a homomorphism because $(a + b) \bmod n = (a \bmod n) + (b \bmod n)$ in $\mathbb{Z}_n$.

4. **Hask** (programming languages with a nice type system, such as Haskell or Scala). Objects: types. Arrows: programs (functions) between types. Example: `add` : $\mathbb{Z} \times \mathbb{Z} \to \mathbb{Z}$. The board notes that the category is **Cartesian closed**: the type of functions $\mathbb{Z} \times \mathbb{Z} \to \mathbb{Z}$ is itself an object, equivalent to $\mathbb{Z} \to (\mathbb{Z} \to \mathbb{Z})$ (currying).

5. **Board games.** A category for each board game (chess, tic-tac-toe). The board only names it ("board game $\to$") without defining the arrows, and I leave it undefined too.

6. **Graphs.** Two constructions:
   - Path(G) for a directed graph G: objects are vertices, arrows $V \to W$ are *paths* from V to W; composition is concatenation of paths, and the identity is the empty path.
   - **Grph:** objects are graphs, arrows are graph homomorphisms (maps f with $f(u) \sim f(v) \iff u \sim v$, as written on the board).

7. **Top.** Objects: topological spaces (sets with a notion of "open set"); arrows: continuous functions.

8. **Vect$_\mathbb{R}$.** Objects: real vector spaces; arrows: linear maps (for example $\mathbb{R}^n \to \mathbb{R}$).

Notes

- The theme of the lesson: the *same* abstract definition covers sets, orders, groups, programs, graphs, spaces and vector spaces. A poset is a category with at most one arrow between any two objects; this is the sense in which "arrows generalize $\leq$".
- At the top left of the board a stray "$x \in y$" is written; it is unrelated to the definitions below it.